

Civil Engineering Level 6

Calculus and Differential Methods

Learner Notes - Professional Equation Edition

These notes support civil engineering learners in differentiation, integration, ordinary differential equations, and numerical methods.

Learning outcome	Learning focus
Perform Calculus Operations	Differentiation and integration of algebraic, trigonometric, logarithmic, and exponential functions.
Solve Ordinary Differential Equations	First-order ODEs, second-order ODEs, and simple engineering models.
Perform Numerical Methods	Roots of equations, interpolation, extrapolation, trapezoidal rule, and Simpson's rule.

1. Importance of Calculus in Civil Engineering

Calculus is used when an engineering quantity changes continuously. Civil engineering technicians apply it in slopes, gradients, rates of change, areas, volumes, flow, settlement, deflection, and numerical approximation.

- Differentiation determines gradients, rates of change, maximum values, and minimum values.
- Integration determines accumulated quantities such as area, volume, total flow, work, and total change.
- Ordinary differential equations model relationships involving rates of change.
- Numerical methods give approximate solutions when exact methods are difficult.

2. Differentiation

The derivative measures the rate at which one quantity changes with respect to another.

$$y = f(x) \Rightarrow \frac{dy}{dx} = f'(x)$$

2.1 Power Rule

$$\text{If } y = ax^n, \quad \frac{dy}{dx} = anx^{n-1}$$

Example: Differentiate the function below.

$$y = 5x^3 - 4x^2 + 7x - 9$$

$$\frac{dy}{dx} = 15x^2 - 8x + 7$$

2.2 Product Rule

$$\text{If } y = uv, \quad \frac{dy}{dx} = u \frac{dv}{dx} + v \frac{du}{dx}$$

Example:

$$y = x^2(3x + 4)$$

$$\frac{dy}{dx} = x^2(3) + (3x + 4)(2x) = 9x^2 + 8x$$

2.3 Quotient Rule

$$\text{If } y = \frac{u}{v}, \quad \frac{dy}{dx} = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

2.4 Chain Rule

$$\text{If } y = f(u) \text{ and } u = g(x), \quad \frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Example:

$$y = (2x + 5)^4$$

$$\frac{dy}{dx} = 4(2x + 5)^3(2) = 8(2x + 5)^3$$

3. Civil Engineering Applications of Differentiation

3.1 Road Profile Gradient

Suppose a road profile is modelled by:

$$y = 0.02x^2 + 0.5x + 100$$

The gradient function is:

$$\frac{dy}{dx} = 0.04x + 0.5$$

At a chainage of 20 m:

$$\left. \frac{dy}{dx} \right|_{x=20} = 0.04(20) + 0.5 = 1.3$$

3.2 Maximum and Minimum Values

Stationary points occur where:

$$\frac{dy}{dx} = 0$$

For the curve below:

$$y = x^2 - 6x + 10$$

$$\frac{dy}{dx} = 2x - 6$$

$$2x - 6 = 0 \Rightarrow x = 3$$

4. Integration

Integration is the reverse process of differentiation. It is used to find accumulated quantities.

$$\int f(x) dx = F(x) + C$$

4.1 Basic Integration Rule

$$\int ax^n dx = \frac{ax^{n+1}}{n+1} + C, \quad n \neq -1$$

Example:

$$\int (6x^2 - 4x + 3) dx = 2x^3 - 2x^2 + 3x + C$$

4.2 Definite Integrals

$$\text{Area} = \int_a^b f(x) dx$$

Example:

$$\int_0^4 (2x + 1) dx = [x^2 + x]_0^4 = 20$$

5. Ordinary Differential Equations

An ordinary differential equation contains derivatives of an unknown function with respect to one independent variable.

$$\frac{dy}{dx} = ky$$

5.1 Separation of Variables

$$\frac{dy}{dx} = xy$$

$$\frac{1}{y} dy = x dx$$

$$\int \frac{1}{y} dy = \int x dx$$

$$\ln y = \frac{x^2}{2} + C$$

$$y = Ae^{x^2/2}$$

5.2 First-Order Linear Differential Equations

$$\frac{dy}{dx} + P(x)y = Q(x)$$

$$IF = e^{\int P(x) dx}$$

5.3 Second-Order Differential Equations

$$a\frac{d^2y}{dx^2} + b\frac{dy}{dx} + cy = f(x)$$

6. Numerical Methods

6.1 Bisection Method

$$C = \frac{a+b}{2}$$

If the signs of the function values change across an interval, a root lies inside that interval.

6.2 Newton-Raphson Method

$$X_{n+1} = X_n - \frac{f(X_n)}{f'(X_n)}$$

6.3 Trapezoidal Rule

$$\int_a^b y dx \approx \frac{h}{2}[y_0 + y_n + 2(y_1 + y_2 + \cdots + y_{n-1})]$$

6.4 Simpson's One-Third Rule

$$\int_a^b y \, dx \approx \frac{h}{3} [y_0 + y_n + 4\sum y_{\text{odd}} + 2\sum y_{\text{even}}]$$

Simpson's one-third rule requires an even number of strips.

7. Practice Questions

1. Differentiate $y = 4x^4 - 3x^2 + 6x - 8$.
2. Differentiate $y = (5x - 2)^3$.
3. Find the stationary point of $y = x^2 - 8x + 15$.
4. Integrate $3x^2 + 2x - 5$ with respect to x .
5. Evaluate the definite integral of x^2 from $x = 0$ to $x = 3$.
6. Use the trapezoidal rule to approximate the area under $y = x^2 + 1$ from $x = 0$ to $x = 4$ using $h = 1$.